MILITARY SPECIFICATION

TRIAMTERENE AND HYDROCHLOROTHIAZIDE CAPSULES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Triamterene and Hydrochlorothiazide Capsules.

1.2 Classification. The capsules covered by this specification shall be the following sizes (see 6.2).

Size 1 - bottles of 100s
Size 2 - bottles of 1000s

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.	(See 6.3)

STANDARDS

Federal

FED-STD-00285	Capsules (for Medicinal Purposes).

FSC 6505

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Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Triamterene and Hydrochlorothiazide Capsules and shall be in accordance with all applicable requirements of FED-STD-00285 and as specified herein:

S2. Classification. Shall be Type I or Type II.

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S5.4 The following is added to this paragraph:

Each capsule shall contain 25 mg of hydrochlorothiazide and 50 mg of triamterene, within the designated assay limits for the capsules.

S6.2.15 Acid solubility. First test paragraph for Type I capsules, delete "5 minutes" and substitute "15 minutes".

3.2 Assay. The capsules shall assay to contain not less than 95.0 percent and not more than 105.0 percent of the required amount of each active ingredient when assayed as specified in 4.3.1.1 for Hydrochlorothiazide and 4.3.1.2 for Triamterene.

3.3 Identification. The active ingredients in the finished capsules shall comply with the identity test for each active ingredient, as follows:

3.3.1 Hydrochlorothiazide. The absorbance curve obtained in the assay method, specified in 4.3.1.1, shall correspond to that obtained from a similarly treated mixture of 1 part by weight of U.S.P. Reference Standard hydrochlorothiazide and 2 parts by weight of triamterene.

3.3.1.1 A copy of the absorbance curve shall be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center for use if samples are submitted for testing.

3.3.2 Triamterene. The absorbance curve obtained in the assay method, specified in 4.3.1.2, shall correspond to that obtained from a U.S.P. Reference Standard.

3.3.2.1 A copy of the standard absorbance curve (graph) shall also be kept on file in the Technical and Quality Assurance Division, Directorate of Medical Materiel, Defense Personnel Support Center for use if samples are submitted for testing.

3.4 Ingredients.

3.4.1 Hydrochlorothiazide. The hydrochlorothiazide used in the manufacture of the finished capsules shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.4.2 Triamterene. The triamterene used in the manufacture of the finished capsules shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

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3.4.3 Other ingredients. All other ingredients entering into the finished capsules shall be U.S.P. or N.F. quality, or if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.5 Expiration dating period. The expiration dating period shall be 60 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-00285.

4.3 Tests.

4.3.1 Assay (filled capsules).

4.3.1.1 Hydrochlorothiazide. The hydrochlorothiazide content in the filled capsules shall be determined as follows:

Reagent.

pH 6.0 Phosphate Buffer. Add 250 ml of 0.2 M KH_2PO_4 (27.218 grams of KH_2PO_4 per 1000 ml of purified water) and 28 ml of 0.2 M NaOH (8 grams of NaOH per 1000 ml of purified water) to a 1000 ml volumetric flask. Dilute to approximately 900 ml, check the pH and adjust, if necessary to pH 6.0 $\pm$ 0.05. Dilute to volume with purified water and mix well.

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Procedure.

Mix the contents of 50 capsules in a glass-stoppered Erlenmeyer flask. Quantitatively transfer an accurately weighed sample of about 140 mg to a 100-ml volumetric flask. Add 5 ml of pH 6.0 Phosphate Buffer, 5 ml purified water and swirl to wet the solids. Add about 5 ml of methanol and again swirl to suspend the solids and dissolve the excipient. Rinse down the neck of the flask with about 50 ml of methanol. Stopper and shake until most of the solids have dissolved. Dilute to volume with methanol and mix well. Allow the flask to stand until the residual insoluble material has settled. Transfer a 5 ml aliquot of the clear supernatant liquid (filter through a cotton pledget on the tip of the pipet, if necessary) to a 250-ml volumetric flask. Dilute to volume with methanol and mix well.

Record the ultraviolet absorption spectrum of this solution from 350 to 240 millimicrons (mm) on a Beckman DB-G Recording Spectrophotometer (or equivalent) using methanol in the reference cell. Record a baseline using the same operating conditions with methanol in both cells. Subtract the absorbance at 250 mm from the absorbance at the maximum (270.5 mm).

Calculations:

$$\frac{Au}{As} \times Cs \times \frac{\text{Av. Net Contents (grams/cap.)}}{\text{Sample Wt. (grams)}} \times \frac{250}{5} \times \frac{100}{25} \times 100 = \text{Percent of required amount of hydrochlorothiazide.}$$

Where:

Au = absorbance of sample solution (270.5 mm - 250 mm)
As = absorbance of standard solution (270.5 mm - 250 mm)
Cs = exact final concentration (about 0.005 mg/ml) of standard

4.3.1.2 Triamterene. The triamterene content in the filled capsules shall be determined as follows:

Procedure.

Mix the contents of 50 capsules in a glass-stoppered Erlenmeyer flask. Quantitatively transfer an accurately weighed sample of about 680 mg to a 1000 ml volumetric flask. Add 20 ml of purified water and heat the flask on a steam bath for 2 minutes, swirling the contents occasionally. Add 400 ml of 2-methoxyethanol and continue to heat for 20 minutes with occasional swirling. Add another 400 ml of 2 methoxyethanol and mix well. Add purified water almost to volume, mix well and cool the flask to room temperature. Dilute to volume with purified water and mix well. Transfer a 10 ml aliquot of this solution to a 250-ml volumetric flask, allowing enough time for the pipet to drain completely. Add 5 ml of 88 percent Formic Acid, dilute to volume with purified water and mix well.

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Record the ultraviolet absorption spectrum of this solution from 400 to 300 millimicrons (mm) on a Beckman DB-G Recording Spectrophotometer (or equivalent) using Solvent Mixture (10 ml of 2-methoxyethanol and 5 ml of 88 percent Formic Acid diluted to 250 ml with purified water) in the reference cell. Record a baseline using the same operating conditions with the Solvent Mixture in both cells. Subtract the absorbance at 395 mm from the absorbance at the maximum (357.5 mm).

Calculations:

$$\frac{Au}{As} \times Cs \times \frac{\text{Av. Net Contents (grams/cap)}}{\text{Sample Wt. (grams)}} \times \frac{1000}{50} \times \frac{250}{10} \times 100 = \text{Percent of required amount of Triamterene}$$

Where:

Au = absorbance of sample solution (395 mm - 357.5 mm)
As = absorbance of standard solution (395 mm - 357.5 mm)
Cs = exact final concentration (about 0.008 mg/ml) of standard

4.4 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

NOTE: Above examination is only applicable to the unit, intermediate and exterior containers.

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5. PREPARATION FOR DELIVERY.

5.1 Preparation For Delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (See 6.1).

5.2.1 Unit of issue. One bottle containing one hundred (100) capsules of size 1, or one thousand (1000) capsules of size 2 as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate containers (all sizes). Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. Closure P, child-proof, as specified in PPP-C-186 may be supplied for size 1.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue of size 1 shall be packaged as specified in 5.2.7.1 of PPP-C-186. Twelve (12) units of issue of size 2 shall be packaged as specified in 5.2.9 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue of size 1 or twenty-four (24) units of issue of size 2 shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue of size 1 shall be packed as specified in 5.4.3.1 of PPP-C-186 or twenty-four (24) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container (Sizes 1 and 2). Each immediate container label for sizes 1 and 2 shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label for sizes 1 and 2 shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

Notes: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied for sizes 1 and 2 the National Stock Number shall appear on the immediate container.

5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-901-0024	TRIAMTERENE AND HYDROCHLOROTHIAZIDE CAPSULES, 100s.
6505-00-901-0043	TRIAMTERENE AND HYDROCHLOROTHIAZIDE CAPSULES, 1000s

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6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1384
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-T-37229 TRIAMTERENE AND HYDROCHLOROTHIAZIDE CAPSULES

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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